Power for Repeated-Measures ANOVA

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Introduction

Repeated-measures designs achieve substantial statistical-power gains over independent-groups designs by exploiting the within-subject correlation across the multiple measurements per subject. Each subject contributes information at every time point or condition, so the within-subject component of variance — typically the smaller of the two variance components — drives the sensitivity of the test. Power calculations for repeated-measures ANOVA must therefore explicitly account for the within-subject correlation; ignoring it leads to substantially over-estimated sample-size requirements and wastes resources, while assuming an unrealistically high correlation under-powers the trial when reality intervenes.

Prerequisites

A working understanding of repeated-measures ANOVA, the sphericity assumption and its diagnostics, Cohen's f as the standardised effect-size measure for ANOVA, and the role of within-subject correlation in design efficiency.

Theory

For a within-subjects design with k measurements per subject, effect size f, and average within-subject correlation ρ, the non-centrality parameter of the omnibus F-test is approximately

$$\lambda = nf^2 \frac{k}{1-\rho},$$

where n is the number of subjects. Power is the tail probability of the non-central F distribution evaluated at the conventional critical value. Higher ρ inflates λ and reduces required sample size; near-zero ρ gives little efficiency advantage over an equivalent independent-groups ANOVA. The Cohen-conventional benchmarks for f are 0.10 (small), 0.25 (medium), and 0.40 (large).

Assumptions

Sphericity holds (equal pairwise variances of differences across time points), residuals are approximately Normal, and the within-subject correlation is reasonably known from pilot data or literature. Sphericity violations require Greenhouse-Geisser or Huynh-Feldt corrections, which slightly inflate the required sample size.

R Implementation

```r
library(WebPower)

wp.rmanova(f = 0.25, ng = 1, nm = 4, nscor = 1,
           alpha = 0.05, power = 0.80, type = 1)

rho_grid <- seq(0.2, 0.9, by = 0.1)
ns <- sapply(rho_grid, function(rho) {
```

```
  wp.rmanova(f = 0.25, ng = 1, nm = 4, nscor = rho,
             alpha = 0.05, power = 0.80, type = 1)$n
})
data.frame(rho = rho_grid, n = round(ns))
```

Output & Results

`wp.rmanova()` returns the required sample size at specified f, k, ρ, and target power. The sensitivity table shows required n falling sharply as within-subject correlation increases — from 54 subjects at $\rho = 0.2$ to only 8 at $\rho = 0.9$. This dramatic dependence makes pre-specification of ρ (and a defensible sensitivity range) essential to honest sample-size planning.

Interpretation

A reporting sentence: "For a within-subjects design with four time points, a medium effect ($f = 0.25$), and assumed within-subject correlation 0.5 from pilot data, $n = 34$ subjects provide 80 % power at $\alpha = 0.05$ for the omnibus F-test under the sphericity assumption. Sensitivity analyses across $\rho \in [0.30, 0.70]$ yield required n from 47 to 23 respectively; the protocol enrols 50 to accommodate the most conservative scenario plus 15 % attrition." Always state ρ and report sensitivity.

Practical Tips

- If sphericity is suspected to fail, inflate the calculated sample size by 10–30 % to compensate for the Greenhouse-Geisser or Huynh-Feldt corrections that will be applied at analysis time; the corrections reduce effective degrees of freedom and therefore reduce achievable power.
- For mixed designs combining between-subjects and within-subjects factors, use `wp.rmanova(type = 2)` or `type = 3` for the appropriate stratum, or use simulation for non-standard designs; classical formulas can be brittle in mixed designs with complex interaction structures.
- Higher numbers of repeated measurements per subject inflate power dramatically because each subject contributes more information; planners often face a trade-off between recruiting more subjects (expensive, slow) and measuring each subject more times (cheap, fast).
- When the primary scientific question is a specific contrast (a linear trend over time, a difference between two conditions), a contrast-specific power calculation typically yields more power than the omnibus F-test and should be used in preference; the omnibus test diffuses power across all interactions.
- For non-Normal outcomes, unbalanced data, or designs with informative dropout, simulation via a linear mixed-effects model (using `simr::powerSim()`) is the gold-standard approach; classical formulas assume a balanced complete-data design and Normality.
- Pilot estimates of ρ have substantial uncertainty in small samples; use the lower end of a reasonable CI on ρ for protective sample-size planning rather than the point estimate.

R Packages Used

`WebPower::wp.rmanova()` for canonical repeated-measures-ANOVA power across one-way and mixed designs; `pwr` family of functions for related ANOVA-power tools; `Superpower::ANOVA_power()` for ANOVA-design power simulation including repeated-measures structures; `simr::powerSim()` for simulation-based power on linear mixed-effects models with arbitrary covariance structure; `afex::aov_car()` for fitting the underlying ANOVA model with sphericity correction.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**

- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation